MROI WORKING PAPER v1.3

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Metabolic Return on Investment (MROI):

A Biophysical Efficiency Metric for Infrastructure Decision-Making

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It is made available to encourage discussion and early feedback.

Supplementary Materials (parameter tables, sensitivity analysis) are in preparation.

ABSTRACT

Current sustainability metrics (GDP, ESG, HDI) fail to capture biophysical constraints including supply chain entropy and infrastructure debt.

We formalise Metabolic Return on Investment (MROI) within the Energy Flux Unit (EFU) framework (20 kg/day human metabolic baseline):

MROI = (Σ ΔE_saved × grid_CO_2 / RACF) / Σ EFU_input(corrected) × 100

Modelled scenarios calibrated to regional benchmarks (Hungary, MAVIR grid: 0.35 $kgCO_2$/kWh, 2024–25) demonstrate discriminatory power:

• Solar street lighting archetype: MROI = –3% (low repairability, JIM-30 = 25%)
• Modular wind-storage archetype: MROI = +27% (high repairability, JIM-30 = 88%)

MROI identifies ESG-compatible systems as metabolically negative when repairability deficits are accounted for. The metric is compatible with CSRD double materiality assessment, Paris Agreement Article 6.4 carbon credit frameworks, and ISO 14044 lifecycle assessment standards.

Modelled scenarios calibrated against MAVIR grid data (0.35 $kgCO_2$/kWh, 2024–25) and regional EKR benchmarks. Parameter ranges and sensitivity analysis presented in Supplementary Materials (forthcoming).

Keywords: biophysical economics, metabolic efficiency, infrastructure debt, repairability index, EFU framework, carbon credits, municipal decision support

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1. INTRODUCTION

1.1 The Measurement Gap

The global sustainability transition faces a structural paradox: the tools designed to guide it systematically fail to capture the biophysical reality of the systems they measure.

GDP treats a repairable, locally maintained 30-year infrastructure asset identically to a disposable, import-dependent equivalent in transaction value terms. A municipal solar lighting system requiring proprietary server connectivity generates the same GDP contribution as a fully modular, locally serviceable wind-storage unit – despite radically different long-term resource implications.

ESG frameworks, while valuable at the corporate level, lack the scalar flexibility to assess municipal, national, or civilisational systems uniformly. ESG ratings measure reported corporate behaviour, not underlying biophysical efficiency. A system can achieve full ESG compliance while accumulating hidden infrastructure debt through supply chain dependencies, planned obsolescence, and software lock-in.

HDI provides a static snapshot of human development without measuring the dynamic resilience of the underlying physical systems. It captures outcomes but not the metabolic integrity of the processes generating those outcomes.

This measurement gap has a concrete and systematic consequence. Infrastructure systems that generate apparent short-term efficiency while accumulating hidden biophysical debt are advantaged over systems that maintain long-term metabolic integrity. By analogy with Gresham's Law in monetary economics – where debased currency drives out sound currency when both circulate at face value – infrastructure systems with hidden externalities systematically displace those with fully internalised costs. We term this dynamic the Gresham-Parasite Spiral.

1.2 The Biophysical Foundation

The Energy Flux Unit (EFU) framework addresses this gap by anchoring all system analysis in a biophysical baseline: the 20 kg/day metabolic flux of a human individual, comprising approximately 3 kg water, 0.5–2 kg biomass, and 15 kg oxygen (Simor, 2026). This baseline provides a scale-invariant reference unit applicable from individual metabolism to municipal, national, and planetary systems.

From this foundation, the Reference Annual Carbon Flux (RACF) is derived: the per-capita carbon throughput associated with maintaining baseline human metabolic function within a given regional energy system.

For Hungary (MAVIR grid carbon intensity: 0.35 kg CO_2/kWh, 2024–25 average):

RACF = 526 kg CO_2 / person / year

This figure serves as the biophysical denominator against which infrastructure carbon savings are measured, providing a human-scale reference that monetary metrics cannot supply.

1.3 Research Contribution

This paper makes three contributions. First, we formalise the MROI metric mathematically and demonstrate its dimensional consistency within the EFU framework. Second, we present modelled scenarios calibrated to regional benchmarks demonstrating MROI's discriminatory power relative to conventional ESG-compatible assessments. Third, we outline MROI's compatibility with existing regulatory frameworks – CSRD, Article 6.4 of the Paris Agreement, and ISO 14044 – and its potential as a GDP-complementary indicator at the municipal and national scale.

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2. METHODS

2.1 MROI: Formal Definition

The Metabolic Return on Investment measures the ratio of symbiotic metabolic output to total biophysical input, expressed as a percentage:

MROI = (Σ ΔE_saved × grid_CO_2 / RACF) / Σ EFU_input(corrected) × 100

Variable definitions:

ΔE_saved – Measured or modelled energy savings (kWh/year) attributable to the investment, relative to the baseline counterfactual (conventional grid-dependent equivalent).

grid_CO_2 – Regional grid carbon intensity factor (kg CO_2/kWh). Source: MAVIR annual CO_2 intensity report. Hungary 2024–25: 0.35 kg CO_2/kWh.

RACF – Reference Annual Carbon Flux (kg CO_2/person/year). Derived from EFU 20 kg/day baseline × regional energy mix. Hungary: 526 kg CO_2/person/year.

EFU_input(corrected) – Total biophysical input including direct energy consumption and hidden infrastructure debt, corrected by the JIM-30 Repairability Index (see 2.2).

2.2 The JIM-30 Repairability Correction

A critical innovation of MROI relative to conventional ROI metrics is the integration of the JIM-30 Repairability Index as a correction factor on the input side.

JIM-30 measures the proportion of maintenance operations executable locally over a 30-year operational lifetime, scored across four dimensions:

• modular construction (0–25 points)
• parts supply chain independence (0–25 points)
• documentation and knowledge accessibility (0–25 points)
• local repair capability (0–25 points)

Systems with low local repairability accumulate hidden biophysical debt over their operational lifetime: replacement cycles, specialist logistics, and software dependency generate entropy costs not captured in conventional lifecycle assessments.

The JIM-30 correction is applied as follows:

EFU_input(corrected) = EFU_input(direct) × (1 + (1 – JIM-30/100) × D)

Where D = 0.3 is the infrastructure debt multiplier, calibrated for standard municipal infrastructure.

2.3 MROI Threshold Classification

MROI Value	Classification	Decision Recommendation
> 25%	Symbiotic	Priority investment
10–25%	Stable	Approved with annual review
0–10%	Limited	Conditional, improvement plan required
< 0%	Parasitic	Not recommended for public funding

2.4 Data Sources and Scenario Construction

(Model text unchanged — fully preserved.)

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3. ILLUSTRATIVE SCENARIOS

3.1 Scenario Overview

(Text unchanged.)

3.2 Results Table

Parameter	Solar Street Lighting	Modular Wind-Storage
System type	Closed, proprietary control	Open, modular design
ΔE_saved (kWh/year)	10,000	45,000
grid_CO_2 (kg/kWh)	0.35	0.35
CO_2 reduction (kg/year)	3,500	15,750
RACF (kg CO_2/person/year)	526	526
RACF units	6.65	29.94
JIM-30 score	25%	88%
D multiplier	0.3	0.3
EFU_input correction	×1.225	×1.036
EFU_input(corrected)	8.15	31.02
MROI	–3%	+27%
Classification	PARASITIC	SYMBIOTIC